MSR113

Where NOT to Use AI in Health Economic Modeling: An Agentic Platform with Clear Trust Boundaries

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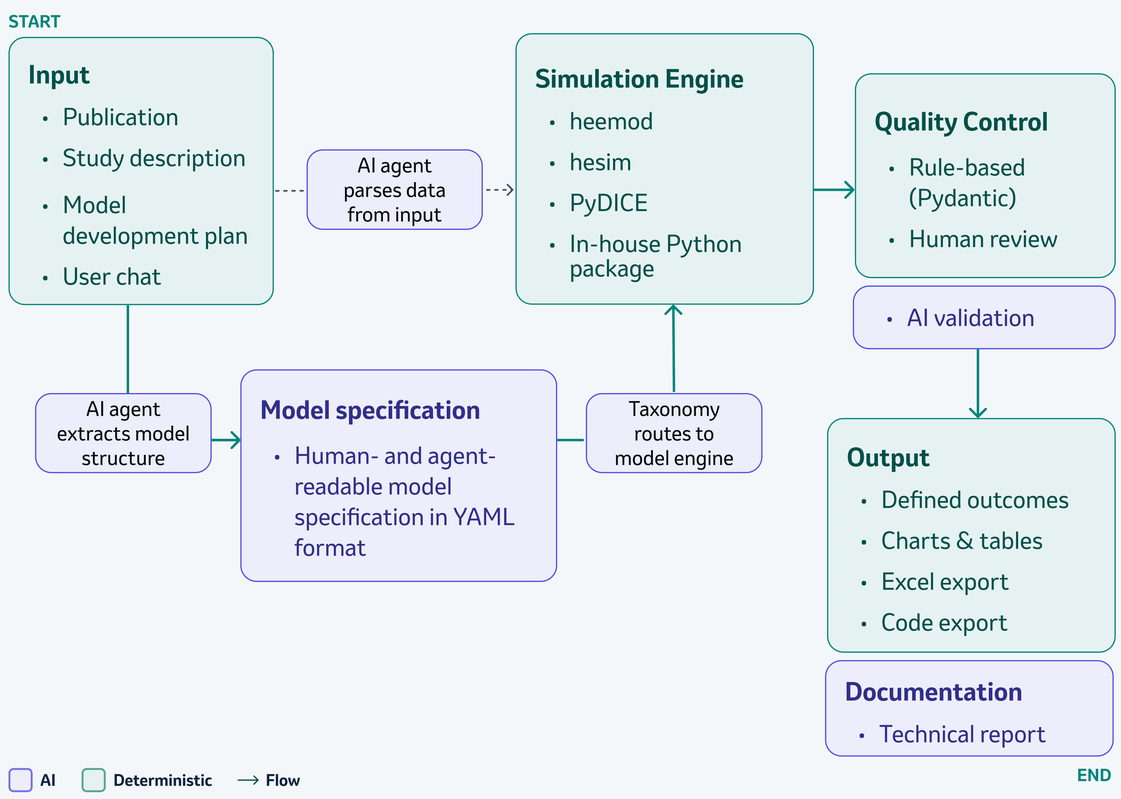
Background & Aims

- Health economists spend considerable time on evidence gathering, iterating model structures, and writing documentation—manual, repetitive tasks well suited to AI.
- However, AI-generated model logic, calculations, and even results are non-deterministic, prone to silent errors, and hard to QC, while HTA bodies expect transparency, reproducibility, and human accountability.
- We developed an AI workflow that accelerates the work around the model, but **never uses AI for computation**: simulation stays in established, deterministic packages.

Methods & Results

- Agents use tools such as *Docling*, which preserves table structure, to extract clinical and economic inputs from publications and HTA reports.
- Extracted inputs and model structure are captured in a human-readable YAML specification that modelers review and approve before simulation. It doubles as documentation and executable input, so changing the model structure (e.g., adding a health state) means editing the file, not recoding.

Fig. 1. AI workflow: AI structures inputs, deterministic engines compute.

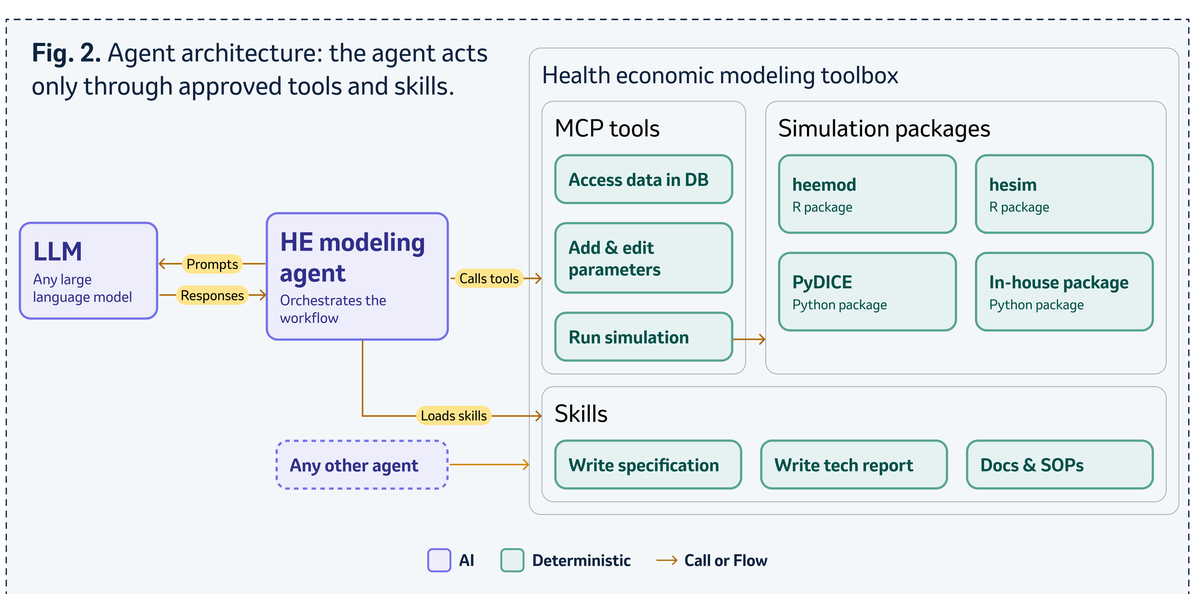


- Simulation is handled by established open-source packages, routed automatically by model taxonomy: cohort state-transition models to *heemod*, individual-level state-transition models to *hesim* (both R), and DICE models to *PyDICE* (Python), or an in-house Python package. The workflow can export executable code (R/Python) or Excel workbooks. All AI-extracted parameters remain traceable to source documents, and results are reproducible through the underlying packages (Fig. 1).

Methods & Results (cont.)

- In a proof-of-concept, the workflow went from a published study or HTA report to a running, reviewable simulation without manual coding (a step that typically takes weeks). It replicated over 90% of the report's original outputs using only the published report, with no access to the underlying calculations.
- The agent works inside a harness: it can act only through approved, quality-controlled tools and skills, exposed via MCP (Model Context Protocol). It calls the engines but **never writes or runs its own simulation code** (Fig. 2).
- Looking ahead, we expect modelers to increasingly work through AI agents built on HEM-specific harnesses: purpose-built MCPs connecting to external data sources and modeling tools, plus lightweight interfaces modelers assemble themselves as needed.

Fig. 2. Agent architecture: the agent acts only through approved tools and skills.



Implications & Conclusions

- We recommend using AI for the work around the model: evidence extraction, structuring, specification, documentation; **deterministic tools handle calculations and simulation code**. The same specification always gives the same results.
- Because every run traces back to a reviewed specification and a QC'd engine, health economists can shift focus from coding to what matters most: model structure and inputs.

Limitations & Next Steps

- Proof-of-concept scope; not yet tested in an HTA submission.
- AI-drafted specifications may differ between runs or contain extraction errors, so human review remains essential before simulation.
- Next, an evidence agent continuously searching for new data could plug into the same system—a stepping stone toward living models.

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